

ORIGINAL ARTICLE

Use of 45-degree implanter angulation with bevel-to-skin orientation to enhance follicular entry

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ABSTRACT

Background: Implanting the follicular units precisely is crucial for the ultimate result of follicular unit extraction (FUE) hair transplantation. The angle of implantation and the position of the implanter bevel are extremely important for reducing follicular trauma and obtaining natural hair direction.

Objective: This study aims to assess whether the use of implanters angled at 45 degrees with bevel-to-skin orientation can improve follicular entry and early clinical outcomes as compared to the traditional perpendicular implantation technique.

Materials and methods: In this randomized, split-scalp clinical trial, FUE hair transplantation patients had their scalp divided into two equal halves. Implantation on the test side was done with implanters angulated at 45 degrees with bevel-to-skin orientation while the control side was implanted with a conventional perpendicular method. Both operative parameters such as ease of follicular entry, graft resistance, follicular popping, and implantation time and early post-operative follicular integrity were evaluated.

Results: The 45-degree angulation method demonstrated superiority in the ease of follicular entry, less graft resistance and follicular popping, as well as a shorter implantation time compared to the conventional method. Early post-operative evaluation showed improved graft stability and more favorable hair angulation on the test side without any increase in complications.

Conclusion: Implanter angulation at 45 degrees with bevel-to-skin orientation successfully improves follicular entry and the overall efficiency of the operation in FUE hair transplantation and may improve early clinical outcomes.

Key words: Implanters, 45 Degrees, Density, FUE



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Introduction

Follicular unit extraction (FUE) hair transplantation is one of the areas in which medical practice has dramatically changed and developed over the last ten years¹. The main goal has been to achieve as natural-looking and long-lasting aesthetic outcomes as possible, with greater emphasis on gentle handling of grafts and exact implantation². Although extraction methods have been improved such that graft quality is higher, the implantation step still has a major influence on follicular survival, hair direction, and hair density³. Therefore, the stress brought by the mechanical method of implanting the graft, the wrong angulation, and the improper orientation of the implanter can all hamper the follicles which may result in the bending, popping or improper growth patterns⁴.

Implanter - based methods were created to ensure uniformity in depth, direction, and density of follicular placement⁵. However, the conventional implantation method is typically performed at near-perpendicular angles which entail higher resistance at the epidermal-dermal interface especially when scalp tissue is firm or fibrotic. The position of the bevel of the implanter in relation to the skin is an important factor that determines how the graft is introduced to the recipient site⁶. It affects tissue displacement, graft alignment and the ease of insertion. Therefore, changing these technical aspects may result in an improvement in follicular penetration as well as a reduction in injury⁷.

An implanter angulated at 45 degrees with bevel-to-skin orientation is hypothesized to facilitate follicular penetration by reducing insertional resistance and aligning the graft with the natural hair exit angle⁸. Despite these theoretical advantages, the lack of quantitative clinical data confirming the effectiveness of this method remains a gap in the literature⁹.

This randomized, split-scalp clinical trial aims at systematically investigating whether the use of this new implantation method in placing follicular units results in higher success in follicular entry, less graft trauma and early postoperative outcomes as compared to the conventional method of perpendicular implantation.

Materials and methods

Study design

This is a randomized, split-scalp clinical trial that was carried out on patients who underwent FUE hair transplantation, following obtainment of informed consent and the institutional ethical approval (SRB/SDC/PhD/OMFS-2343/23/TH/-111). Adult patients who were planned for FUE hair transplantation and had similar scalp characteristics on both sides of the recipient area were included. Patients who had active scalp infections, cicatricial alopecia, systemic conditions that could negatively affect wound healing and patients who had previous hair transplantation in the recipient area were not included.

Surgical protocol

Each patient underwent the standard FUE graft extraction procedure from the back of the head (occipital area) with the same uniform extraction technique to maintain graft quality as consistently as possible. The harvested follicular units were kept in a cold isotonic solution until the time of implantation. To avoid patient dependent effects, a split scalp design was chosen and the recipient main scalp was divided for each patient into two equal halves through the midline. Computer generated randomization was done to decide which side would be the test area and which would be the control area.

Follicular units on the test side were implanted through implanters inclined at 45 degrees to the scalp surface with the bevel facing the skin (Figure 1). On the control side, the implantation was carried out by the conventional method of perpendicular implanter angulation with standard bevel orientation. To exclude any inter-operative variations, the same surgeon implanted all the grafts using the same implanter sizes and graft densities at the same time.

Outcome measures and data collection

Parameters assessed intraoperatively included ease of follicular entry, resistance during insertion, occurrence of graft popping or bending and time taken



Figure 1. Hair Grafts implanted at 45 degrees.

for implantation per graft. Immediate post-implantation evaluation was based on follicular integrity and graft stability. Patients were followed up postoperatively at scheduled intervals to monitor early healing and graft retention. Data were collected for all observations and analyzed statistically using SPSS Statistics Version 28.0 (IBM Corp., Armonk, NY) to determine differences in outcomes between the two methods.

Results

A total of 30 patients were included in the study, contributing an equal number of grafts to both the test and control sides using the split-scalp design. A total of 3,000 follicular units were extracted, with 1,500 grafts implanted using the 45-degree angulation with bevel-to-skin orientation (test side - Figure 2) and 1,500 grafts using the conventional perpendicular technique (control side - Figure 3). Baseline scalp characteristics and graft distribution were comparable between both sides.



Figure 2. Test group with grafts implanted at 45 degrees.

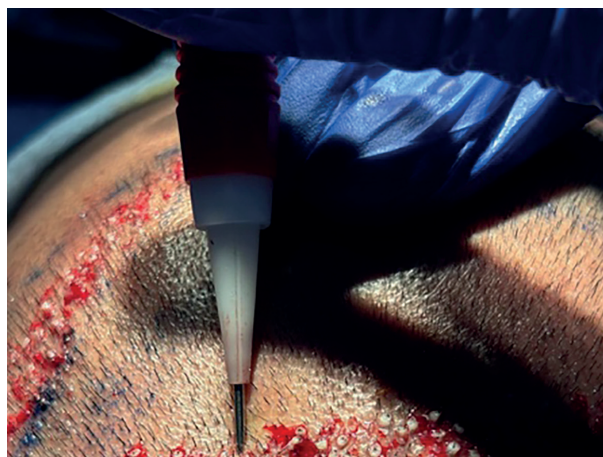


Figure 3. Control group with regular implantation.

Intraoperatively, the test side demonstrated a significantly smoother follicular entry with reduced resistance at the recipient site. The incidence of graft popping, and visible follicular bending was noticeably lower with the 45-degree angulated technique. Implantation efficiency was improved, reflected by a reduced mean time per graft (Table 1). Immediate post-implantation assessment showed superior graft stability and alignment on the test side, with follicles lying more closely along the natural hair exit angle. No intraoperative or early postoperative complications were observed on either side (Table 2).

Early postoperative evaluation revealed better graft retention and less follicular trauma on the test side. Healing was comparable between both

Table 1. Intraoperative Graft-Related Outcomes.

Parameter	Test Side (45° Angulation)	Control Side (Perpendicular)
Total grafts implanted	1,500	1,500
Mean implantation time per graft (seconds)	6.8 ± 0.9	8.1 ± 1.1
Graft popping (%)	3.2%	9.6%
Graft bending (%)	2.8%	8.4%
Ease of follicular entry (subjective score)	High	Moderate

Table 2. Immediate Post-Implantation Assessment.

Parameter	Test Side	Control Side
Intact follicular units (%)	97.1%	90.5%
Graft stability	Good	Moderate
Alignment with natural hair angle	Favorable	Less favorable
Immediate complications	Nil	Nil

techniques, with no statistically significant difference in erythema, edema, or infection rates.

Overall, the use of implanters angulated at 45 degrees with bevel-to-skin orientation resulted in superior intraoperative handling of grafts, reduced mechanical trauma, and improved early graft stability compared with the conventional perpendicular implantation technique.

Discussion

In comparison to traditional near-perpendicular placement, our randomized split-scalp trial demonstrated that implanting follicular units with implanters held at a 45° angulation and the bevel oriented toward the skin improved intraoperative handling (smoother entry, less resistance), decreased rates of graft popping and bending, shortened implantation time per graft, and produced better immediate graft alignment¹⁰. These results lend credence to the theory that the mechanical interface between the graft, needle, and recipient tissue can be significantly altered by a purposeful change in instrument angle and bevel orientation. Such improvements are important for early graft stability and the surgeon's ability to place dense, naturally angled hair⁵.

The graft can seat with less tissue displacement and less spring-back from the wound edges when the implanter is angled at a 45° angle because this reduces the effective shear and compressive forces at the epidermal–dermal junction during entry¹¹. The tendency for the wound edges to push the graft back out (popping) or to misalign the follicle (bending) can be reduced by orienting the bevel toward the skin, which can produce a smoother cutting/parting action or better accommodation of the graft within the lumen during insertion¹². In order to facilitate easier introduction and controlled graft placement, practical guides and technique papers on sharp and dull needle implanter methods place a strong emphasis on hand/wrist positioning (often with an internal wrist angle of about 45°) and thoughtful bevel choices¹³.

Our findings are consistent with the literature that reports quicker implantation, greater ease, and good early outcomes with implanter-based techniques as compared to free placement methods or insertion into premade slits, although there are very few direct comparisons of specific angulations and bevel orientations¹⁴. Various clinical and technique reports have emphasized that the use of the implanter significantly increases the efficiency of the procedure, and allows for dense packing when the technique is carefully employed¹⁵. On the other hand, earlier reviews and

experimental works have sounded a note of caution: only a handful of studies indicate that a more perpendicular graft placement might be linked to better survival isolated instances, presumably because the orientation of the graft with respect to the vascular bed and extent of tissue injury due to shearing are complex and multifactorial phenomena¹⁶. This means that a 45-degree bevel-to-skin angle not only enhances handling and immediate graft seating, but also its effect on long-term survival should be considered in the context of other factors (slit depth, graft fit, vascularity, patient skin properties).

Clinical implications and future directions: A straightforward, inexpensive technical modification that seems to increase implantation efficiency and lessen immediate mechanical graft issues (popping, bending) is the 45° angulation with bevel-to-skin orientation. Surgeons may use this orientation as part of a protocol that still prioritizes appropriate slit design, hemostasis, and gentle graft handling in order to speed up operations and enhance early graft stability¹⁷. Future research should: track cohorts for a full year or longer to evaluate survival and cosmetic results; compare this method for various scalp biotypes and when using premade slits versus simultaneous incision/implantation; and assess learning curves among technicians and assistants who frequently carry out implantation in high-volume settings.

The results show that a straightforward change in implanter angulation and bevel orientation can significantly improve graft handling and early clinical outcomes, even though the current study was primarily concerned with intraoperative and early outcomes. To determine whether these early advantages result in long-term gains in hair growth and patient satisfaction, long-term follow-up studies assessing graft survival, density, and aesthetic outcomes are necessary.

Our findings indicate that a 45° implanter angulation with bevel-to-skin orientation provides observable intraoperative and early postoperative advantages over traditional perpendicular placement; however, more prospective research is needed to determine the ultimate impact on long-term graft survival and patient-reported aesthetic outcomes.

Conclusion

Using implanters angulated at 45-degrees with a bevel-to-skin orientation offers definite intraoperative advantages over the traditional perpendicular implantation technique in FUE hair transplantation, as demonstrated by this randomized split-scalp clinical trial. The modified method greatly decreased the frequency of graft popping and follicular bending, improved follicular entry, and decreased insertional resistance. With shorter implantation times per graft and improved immediate graft stability and alignment, these advancements resulted in increased operative efficiency.

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Ethics Approval and Consent to Participate: The study got approval from the Institutional Human Ethical Committee (IHEC) with the reference number – IHEC/SDC/PhD/OMFS-2343/23/TH-092. Treatment consents were obtained from all participants after explaining the procedure to the patient.

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